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Author(s): Livescu, Daniel
Zi, Zhaorui
Chapurin, Oleksandr
Brady, Peter T.

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Generalized Cahn-Hilliard Navier-Stokes Equations for Complex Multiphase Flows

Daniel Livescu¹, Z. Li²,
O. Chapurin¹, P. Brady¹

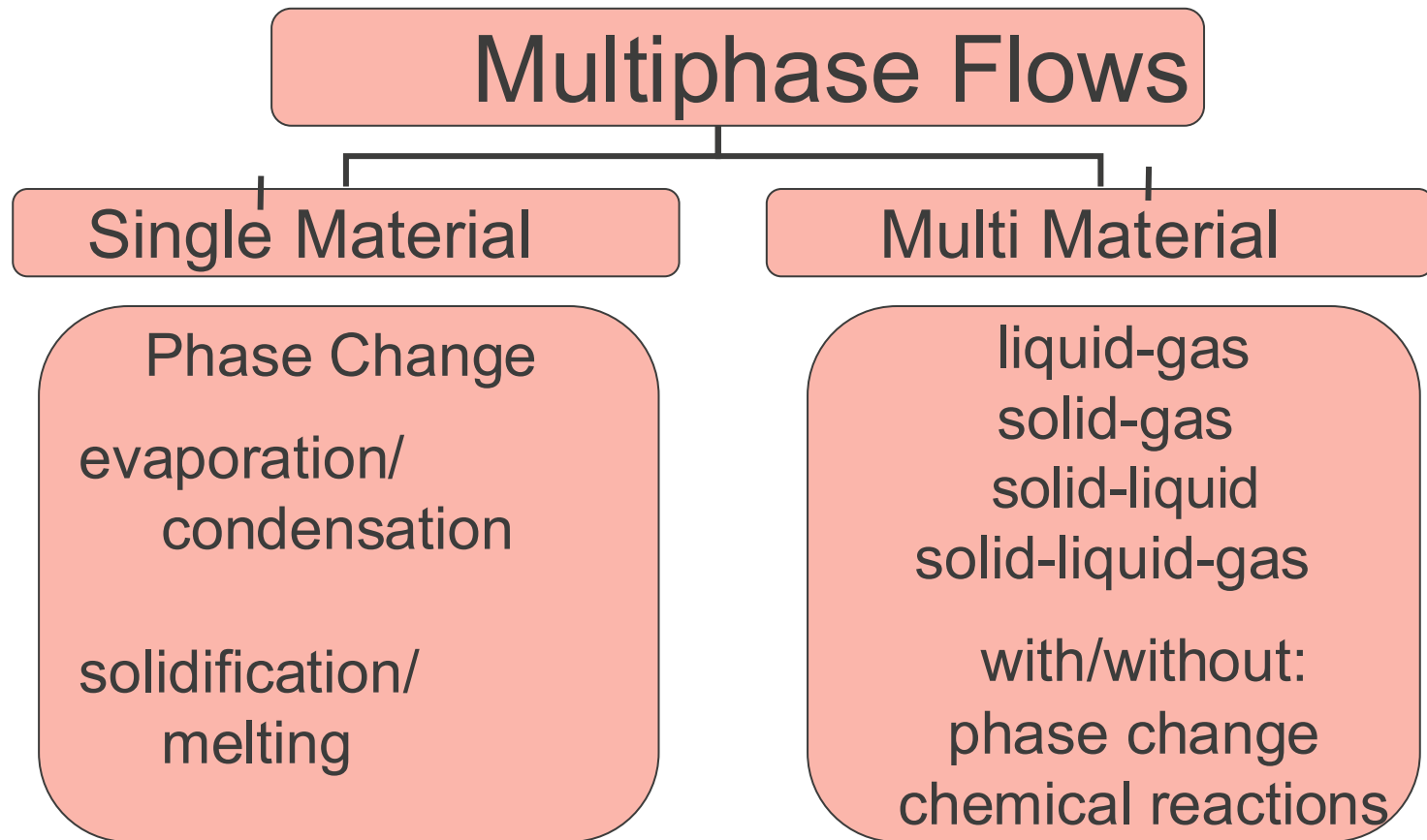
¹ Los Alamos National Laboratory

² Texas A&M Corpus Christi

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Generalized Cahn-Hilliard Navier-Stokes Equations for Complex Multiphase Flows



Our focus is on liquid-gas systems. Solid phase can be added with different methods.

Comparison with other methods

❖ Current numerical approaches and their limitations

➤ Sharp Interface Methods (zero interface thickness):

- Front tracking
- Volume of Fluid (VOF)
- Level Set Method

Problems: A) Ad-hoc formulation for interface properties, B) Difficulties handling changes in surface topology, C) Difficulties satisfying physical constraints and even basic conservation.

➤ Diffuse Interface Methods (non-zero interface thickness)

- Conventional Cahn-Hilliard (C-H) Method

Problems: A) Only physically derived for the incompressible constant density binary case; B) interface is artificially thicker than physical interface.

➤ Non-Continuum Methods: different degrees of fidelity, with atomistic simulations fully capturing the physics of the interface.

❖ Current work: Generalized Cahn-Hilliard Navier-Stokes equations

- Mathematical derivation of general equations (**physically consistent**) describing multi-component ($N \geq 2$) compressible flows from basic thermodynamics (**CGCHNS-Compressible Generalized Cahn-Hilliard Navier-Stokes** equations).
- Derivation of general equations describing multi-component ($N \geq 2$) incompressible flows by taking the incompressible limit of the compressible equations (**IGCHNS-Incompressible Generalized Cahn-Hilliard Navier-Stokes** equations).
- Both CGCHNS and IGCHNS equations can describe flows with arbitrary density ratios and naturally handle complex topological changes of the interface.

Start with general conservation equations without the specification of the transport terms.

❖ General Conservation Equations for Multi-Component Flows

$$\text{Continuity : } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{U}) = 0$$

$$\text{Momentum : } \frac{\partial}{\partial t} (\rho \vec{U}) + \nabla \cdot (\rho \vec{U} \otimes \vec{U}) = \nabla \cdot \underline{\underline{\sigma}}^r + \nabla \cdot \underline{\underline{\sigma}}^v + \sum_{\alpha=1}^N n_{\alpha} \vec{F}_{\alpha}$$

$$\text{Internal energy : } \frac{\partial}{\partial t} (\rho e) + \nabla \cdot (\rho e \vec{U}) = \underline{\underline{\sigma}}^r : \nabla \vec{U} + \underline{\underline{\sigma}}^v : \nabla \vec{U} - \nabla \cdot \vec{q} + \dot{Q} + \rho \sum_{\alpha=1}^N n_{\alpha} \vec{V}_{\alpha} \vec{F}_{\alpha}$$

$$\text{Species mass fraction : } \frac{\partial}{\partial t} (\rho Y_{\alpha}) + \nabla \cdot (\rho Y_{\alpha} \vec{U}) = -\nabla \cdot \vec{J}_{\alpha} + \dot{w}_{\alpha} W_{\alpha}; \quad \vec{J}_{\alpha} = \rho Y_{\alpha} \vec{V}_{\alpha}$$

$$\text{(Species molar concentration) : } \frac{\partial}{\partial t} (n_{\alpha}) + \nabla \cdot (n_{\alpha} \vec{U}) = -\nabla \cdot \left(n_{\alpha} \vec{V}_{\alpha} \right) + \dot{w}_{\alpha}$$

$$\text{Entropy : } \frac{\partial}{\partial t} (\rho s) + \nabla \cdot (\rho s \vec{U}) = -\nabla \cdot \vec{q}_s + \Delta_s$$

where $\vec{J}_{\alpha}$ is the mass flux, $\vec{V}_{\alpha}$ is the diffusion velocity, W_{α} is the molar mass, n_{α} is the molar concentration of species α ($n_{\alpha} = \frac{\rho Y_{\alpha}}{W_{\alpha}} = \frac{\# \text{ moles}}{\text{volume}}$), Y_{α} is the mass fraction of species α , $\underline{\underline{\sigma}}^r$ is the reversible stress, $\underline{\underline{\sigma}}^v$ is the viscous stress, $\vec{q}$ is the heat flux, Δ_s is the entropy source, $\vec{q}_s$ is the entropy flux, $\vec{F}_{\alpha}$ is the external force per unit mole of substance α .

Consider the Helmholtz free energy within a fluid element small compared to the width of the interface.

- ❖ Write the Helmholtz free energy as a sum of attractive and repulsive contributions:

$$F = F^A + F^R$$

- ❖ Assume that the attractive part also depends on gradients of the concentration (original C-H assumption).

- ❖ Assume continuum hypothesis holds across the interface, i.e. the fluid element much smaller than interface thickness.

- ❖ Start from the Helmholtz free energy per unit volume $\hat{F}$ (since we are interested in transport equations) and expand around $\nabla^p n_\alpha = 0$, retaining terms to second order.

$$\hat{F} = \hat{F}^0(T, n_\alpha) + \hat{F}^I(T, n_\alpha) + \sum_{\alpha=1}^N \gamma_\alpha(T, n_\alpha) \nabla^2 n_\alpha + \frac{1}{2} \sum_{\alpha=1}^N \sum_{\beta=1}^N \lambda_{\alpha\beta}(T, n_\alpha) \nabla n_\alpha \cdot \nabla n_\beta$$

Bulk contribution, related to the attractive potential between like-molecules

Interface contribution, related to the *repulsive potential* between the immiscible components

Non-local contributions (Cahn-Hilliard), related to the *attractive potential* between like-molecules.

General or extended thermodynamics accounting for the interface.

- ❖ **Neglect second derivatives in the expansion for Helmholtz free energy** (since the fluid element is much smaller than the interface width, the interface can be assumed locally linear).
- ❖ **Consider ∇n as an extra intensive variable** (if the fluid element doubles in size, the interface remains the same, so the concentration gradient does not change):

Internal energy:

$$\hat{E} = \hat{E}(\hat{S}, n_{1,2,\dots,N}, \nabla n_{1,2,\dots,N}) \quad d\hat{E} = Td\hat{S} + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}) + \sum_{\alpha=1}^N \vec{\Phi}_{\alpha} d(\nabla n_{\alpha})$$

Helmholtz free energy:

$$\hat{F} = \hat{F}(T, n_{1,2,\dots,N}, \nabla n_{1,2,\dots,N}) \quad d\hat{F} = -\hat{S}dT + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}) + \sum_{\alpha=1}^N \vec{\Phi}_{\alpha} d(\nabla n_{\alpha})$$

Gibbs free energy:

$$\hat{G} = \hat{G}(T, p, n_{1,2,\dots,N}, \nabla n_{1,2,\dots,N}) \quad d\hat{G} = -\hat{S}dT + dp + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}) + \sum_{\alpha=1}^N \vec{\Phi}_{\alpha} d(\nabla n_{\alpha})$$

Enthalpy:

$$\hat{H} = \hat{H}(\hat{S}, p, n_{1,2,\dots,N}, \nabla n_{1,2,\dots,N}) \quad d\hat{H} = Td\hat{S} + dp + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}) + \sum_{\alpha=1}^N \vec{\Phi}_{\alpha} d(\nabla n_{\alpha})$$

General or extended thermodynamics accounting for the interface.

- Define the energy per unit mass by $e\rho = \hat{E}$ and similarly for the other free energies. After taking the differential and using the integral relations, one obtains the differential forms **per unit mass** as:

$$\begin{aligned}
 e &= e(s, \rho, n_{1,2,\dots,N}/\rho, \nabla n_{1,2,\dots,N}) & de &= Tds + \frac{p}{\rho^2} d\rho + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}/\rho) + \sum_{\alpha=1}^N \vec{\varphi}_{\alpha} d(\nabla n_{\alpha}) \\
 f &= f(T, \rho, n_{1,2,\dots,N}/\rho, \nabla n_{1,2,\dots,N}) & df &= -sdT + \frac{p}{\rho^2} d\rho + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}/\rho) + \sum_{\alpha=1}^N \vec{\varphi}_{\alpha} d(\nabla n_{\alpha}) \\
 g &= g(T, p, n_{1,2,\dots,N}/\rho, \nabla n_{1,2,\dots,N}) & dg &= -sdT + \frac{dp}{\rho} + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}/\rho) + \sum_{\alpha=1}^N \vec{\varphi}_{\alpha} d(\nabla n_{\alpha}) \\
 h &= h(s, p, n_{1,2,\dots,N}/\rho, \nabla n_{1,2,\dots,N}) & dh &= Tds + \frac{dp}{\rho} + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}/\rho) + \sum_{\alpha=1}^N \vec{\varphi}_{\alpha} d(\nabla n_{\alpha})
 \end{aligned}$$

Extended thermodynamic variables resulting from including the interface

⇒ Comparing the differential form of f with the expansion of f (or $\hat{F}$) gives the forms of the **chemical potential**, μ_α , **extended chemical potential**, $\vec{\varphi}_\alpha$, and **pressure**, p :

$$\begin{aligned}\mu_\alpha &= \left. \frac{\partial f}{\partial (n_\alpha/\rho)} \right|_{\rho, T, \frac{n_{\beta \neq \alpha}}{\rho}, \nabla n_\gamma} = \left. \frac{\partial \hat{F}}{\partial n_\alpha} \right|_{T, n_{\beta \neq \alpha}, \nabla n_\gamma} = \mu_\alpha^0 + \mu_\alpha^I + \frac{\rho}{2} \sum_{\beta=1}^N \sum_{\delta=1}^N \left. \frac{\partial \lambda_{\beta\delta}}{\partial n_\alpha} \right|_{\rho, T, n_{\gamma \neq \alpha}} \nabla n_\beta \cdot \nabla n_\delta \\ &= \mu_\alpha^0 + \mu_\alpha^I + \mu_\alpha^1\end{aligned}$$

$$\vec{\varphi}_\alpha = \sum_{\beta=1}^N \lambda_{\alpha\beta} \nabla n_\beta$$

$$p = \rho^2 \left. \frac{\partial f}{\partial \rho} \right|_{T, \frac{n_\alpha}{\rho}, \nabla n_\beta} = p^0 + p^I + \frac{\rho^2}{2} \sum_{\alpha=1}^N \sum_{\beta=1}^N \left. \frac{\partial \lambda_{\alpha\beta}}{\partial \rho} \right|_{T, \frac{n_\delta}{\rho}} \nabla n_\alpha \cdot \nabla n_\beta$$

General forms for the entropy flux, entropy production and reversible stress accounting for the interface

❖ General or Extended Thermodynamics

$$e = e(s, \rho, n_{1,2,\dots,N}/\rho, \nabla n_{1,2,\dots,N}) \quad de = Tds + \frac{p}{\rho^2} d\rho + \sum_{\alpha=1}^N \mu_{\alpha} d(n_{\alpha}/\rho) + \sum_{\alpha=1}^N \vec{\varphi}_{\alpha} d(\nabla n_{\alpha})$$

- Replace $\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \vec{U} \cdot \nabla \rho$, $d\epsilon/dt$, $d(\frac{n_{\alpha}}{\rho})$, $d(\nabla n_{\alpha})$ and ds/dt using continuity, energy, scalar and entropy equations into the above differential equation for de .

Finally, we have:

$$\begin{aligned} \vec{q}_s &= \frac{1}{T} \left(\vec{q} - \sum_{\alpha=1}^N \mu_{\alpha}^* n_{\alpha} \vec{V}_{\alpha} - \sum_{\alpha=1}^N \rho \vec{\varphi}_{\alpha} (n_{\alpha} \nabla \cdot \vec{U} + \nabla \cdot (n_{\alpha} \vec{V}_{\alpha}) - \dot{\omega}_{\alpha}) \right) \\ \Delta_s &= \left(\vec{q} - \sum_{\alpha=1}^N \mu_{\alpha}^* n_{\alpha} \vec{V}_{\alpha} - \sum_{\alpha=1}^N \rho \vec{\varphi}_{\alpha} (n_{\alpha} \nabla \cdot \vec{U} + \nabla \cdot (n_{\alpha} \vec{V}_{\alpha}) - \dot{\omega}_{\alpha}) \right) \cdot \nabla \left(\frac{1}{T} \right) - \sum_{\alpha=1}^N \left(\frac{\nabla \mu_{\alpha}^*}{T} - \frac{\vec{F}_{\alpha}}{T} \right) \cdot \frac{\vec{J}_{\alpha}}{W_{\alpha}} \\ &\quad + \frac{1}{T} \underline{\underline{\sigma}}^v : \nabla \vec{U} - \frac{1}{T} \sum_{\alpha=1}^N \mu_{\alpha}^* \dot{\omega}_{\alpha} + \frac{\dot{Q}}{T} \\ \underline{\underline{\sigma}}^r &= \left(-p + \sum_{\alpha=1}^N n_{\alpha} \nabla \cdot (\rho \vec{\varphi}_{\alpha}) \right) \underline{\underline{I}} + \sum_{\alpha=1}^N (\nabla n_{\alpha} \otimes \rho \vec{\varphi}_{\alpha}) \end{aligned}$$

where $\mu_{\alpha}^* = \mu_{\alpha} - \nabla \cdot (\rho \vec{\varphi}_{\alpha})$ is the **reduced generalized chemical potential**

Combine the second law with Onsager relation for irreversible processes

- The second law requires $\Delta_s \geq 0$. The viscous contribution satisfies this relation, and we have proven it for the reaction term under general conditions (Standridge, Livescu, Cizmas, 2026). In compact form, the rest of the contributions can be written as:

$$\vec{J}_q \cdot \left(-\frac{\nabla T}{T}\right) + \sum_{\alpha=1}^N \vec{J}_\alpha \cdot (-\Lambda_\alpha) \geq 0, \text{ where:}$$

$$\vec{J}_q = \vec{q} - \sum_{\alpha=1}^N H_\alpha \frac{\vec{J}_\alpha}{W_\alpha} - \sum_{\alpha=1}^N (\mu_\alpha^I + \mu_\alpha^1) \frac{\vec{J}_\alpha}{W_\alpha} + \sum_{\alpha=1}^N \nabla \cdot \left(\sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_\beta \right) \frac{\vec{J}_\alpha}{W_\alpha} - \sum_{\alpha=1}^N \sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_\beta \left(n_\alpha \nabla \cdot \vec{U} + \nabla \cdot \frac{\vec{J}_\alpha}{W_\alpha} - \dot{\omega}_\alpha \right)$$

is the generalized heat flux and

$$\vec{\Lambda}_\alpha = \frac{1}{W_\alpha} \left(\sum_{\beta=1}^N \frac{\partial \mu_\alpha^0}{\partial x_\beta} \Big|_{p_0, x_{\delta \neq \alpha, \beta}, \nabla x_\delta} \nabla x_\beta + \bar{V}_\alpha \nabla p^0 + \nabla (\mu_\alpha^I + \mu_\alpha^1) - \nabla \left(\nabla \cdot \left(\sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_\beta \right) \right) - \vec{F}_\alpha \right)$$

- ❖ **General irreversible solutions to the inequality equation (*Onsager relations*)** using fluxes and their affinities as $\vec{J}_\alpha$, $-\vec{\Lambda}_\alpha$ and $\vec{J}_q$, $-\frac{\nabla T}{T}$. Since $\vec{J}_\alpha$ are not linearly independent:

$$\vec{J}_\alpha = -a_{0\alpha} \left(\frac{\nabla T}{T} \right) - \sum_{k=1, k \neq j}^N a_{\alpha k} (\vec{\Lambda}_k - \vec{\Lambda}_j)$$

$$\vec{J}_q = -a_{00} \left(\frac{\nabla T}{T} \right) - \sum_{k=1, k \neq j}^N a_{k0} (\vec{\Lambda}_k - \vec{\Lambda}_j)$$

Onsager relations give the mass and heat fluxes

- Onsager's reciprocal relations: $a_{\alpha\beta} = a_{\beta\alpha}$
- Extend the reciprocal relations by defining: $a_{jk} = a_{kj} = -\sum_{i=1, i \neq j}^N a_{ik} \quad \forall j = 1, \dots, N$ to eliminate the arbitrary j term. Then by construction $\sum_{i=1}^N a_{ik} = 0 \quad \forall k = 1, \dots, N$
- Then:

$$\begin{aligned}\vec{J}_\alpha &= -a_{0\alpha} \left(\frac{\nabla T}{T} \right) - \sum_{k=1}^N a_{\alpha k} \vec{\Lambda}_k \\ \vec{J}_q &= -a_{00} \left(\frac{\nabla T}{T} \right) - \sum_{k=1}^N a_{k0} \vec{\Lambda}_k\end{aligned}$$

- Additional constraints on the coefficients to satisfy the second law:

$$a_{00} + \sum_{\alpha=1}^{N-1} \frac{a_{\alpha 0}^2}{a_{\alpha N}} > 0$$

$$\begin{aligned}a_{\alpha k} &< 0, & \forall \alpha \neq k, \alpha = 1, \dots, N; k = 1, \dots, N \\ a_{\alpha\alpha} &> 0, & \forall \alpha = 1, \dots, N\end{aligned}$$

Change in variables for the fluxes' representation

- The linear relations can be defined using a modified set of affinities with the property $\sum_{\alpha=1}^N \vec{d}_{\alpha} = 0$:

$$\begin{aligned} \vec{d}_{\alpha} &= \frac{x_{\alpha} W_{\alpha}}{\mathcal{R} T} (\vec{\Lambda}_{\alpha} - C_p \nabla p + C_F + C_{\mu} + C_{\varphi}) \\ &= \frac{x_{\alpha}}{\mathcal{R} T} \left(\sum_{\beta=1}^N \frac{\partial \mu_{\alpha}}{\partial x_{\beta}} \Big|_{p^0, T, x_{\delta \neq \alpha, \beta}} \nabla x_{\beta} + \left(\bar{V}_{\alpha} - \frac{W_{\alpha}}{\rho} \right) \nabla p^0 - \left(\vec{F}_{\alpha} - \frac{W_{\alpha}}{\rho} \sum_{\beta=1}^N n_{\beta} \vec{F}_{\beta} \right) + \nabla \left(\mu_{\alpha}^I + \mu_{\alpha}^1 - \nabla \cdot \left(\sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_{\beta} \right) \right) \right. \\ &\quad \left. - \sum_{\beta=1}^N \frac{x_{\beta} W_{\alpha}}{W} \nabla \left(\mu_{\beta}^I + \mu_{\beta}^1 - \nabla \cdot \left(\sum_{\gamma=1}^N \rho \lambda_{\beta\gamma} \nabla n_{\gamma} \right) \right) \right) \end{aligned}$$

- Then:

$$\begin{aligned} \vec{J}_{\alpha} &= -a_{0\alpha} \left(\frac{\nabla T}{T} \right) - \mathcal{R} T \sum_{k=1}^N \frac{a_{\alpha k}}{x_k W_k} \vec{d}_k \\ \vec{J}_q &= -a_{00} \left(\frac{\nabla T}{T} \right) - \mathcal{R} T \sum_{k=1}^N \frac{a_{k0}}{x_k W_k} \vec{d}_k \end{aligned}$$

- In principle, the Onsager coefficients can be calculated directly from molecular dynamics.

Summary of the results for the transport terms

❖ Neglect Soret and Dufour effects ($a_{0\alpha} = 0$)

$$\vec{J}_\alpha = - \sum_{\beta=1}^N \frac{a_{\alpha\beta}}{W_\beta} \left(\sum_{\delta=1}^N \frac{\partial \mu_\beta}{\partial x_\delta} \Big|_{p^0, T, x_{\gamma \neq \delta, \beta}} \nabla x_\delta + \bar{V}_\beta \nabla p^0 - \vec{F}_\alpha + \nabla \left(\mu_\beta^I + \mu_\beta^1 - \nabla \cdot \left(\sum_{\delta=1}^N \rho \lambda_{\beta\delta} \nabla n_\delta \right) \right) \right)$$

$$\begin{aligned} \vec{q} = & -a_{00} \left(\frac{\nabla T}{T} \right) + \sum_{\alpha=1}^N H_\alpha \frac{\vec{J}_\alpha}{W_\alpha} + \sum_{\alpha=1}^N (\mu_\alpha^I + \mu_{1\alpha}) \frac{\vec{J}_\alpha}{W_\alpha} - \sum_{\alpha=1}^N \nabla \cdot \left(\sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_\beta \right) \frac{\vec{J}_\alpha}{W_\alpha} \\ & + \sum_{\alpha=1}^N \sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_\beta \left(n_\alpha \nabla \cdot \vec{U} + \nabla \cdot \frac{\vec{J}_\alpha}{W_\alpha} - \dot{\omega}_\alpha \right) \end{aligned}$$

$\underline{\underline{\sigma^r}}$

$$= - \left(p^0 + p^I + \frac{\rho^2}{2} \sum_{\alpha=1}^N \sum_{\beta=1}^N \frac{\partial \lambda_{\alpha\beta}}{\partial \rho} \nabla n_\alpha \cdot \nabla n_\beta - \sum_{\alpha=1}^N n_\alpha \nabla \cdot \left(\sum_{\beta=1}^N \rho \lambda_{\alpha\beta} \nabla n_\beta \right) \right) \underline{\underline{I}} - \sum_{\alpha=1}^N \sum_{\beta=1}^N \rho \lambda_{\alpha\beta} (\nabla n_\alpha \otimes \nabla n_\beta)$$

where $H_\alpha = \mu_\alpha^0 - T \frac{\partial \mu_\alpha^0}{\partial T} \Big|_{p_0, x_{\beta \neq \alpha}}$ is the partial molal enthalpy, $\bar{V}_\alpha = \frac{\partial \mu_\alpha^0}{\partial p_0} \Big|_{T, x_{\beta \neq \alpha}}$ is the partial molal volume for the background state and x_α is the mole fraction.

Simplification of the general transport terms

❖ Assume:

1) The tensor $\lambda_{\alpha\beta}$ is diagonal and $\lambda_{\alpha\alpha} = \lambda_\alpha$; λ_α does not depend on n_α so that $\mu_\alpha^1 = 0$.

2) No external forces

3) Background flow is an ideal mixture ($\frac{\partial \mu_\alpha^0}{\partial x_\beta} |_{p^0, T, x_{\delta \neq \alpha, \beta}} = \mathcal{R} \frac{T}{x_\alpha}$ for $\beta = \alpha$ and 0 otherwise; $\frac{\partial \mu_\alpha^0}{\partial p^0} |_{T, x_{\delta \neq \alpha, \beta}}$

$$\mathcal{R} \frac{T}{p}; \mu_\alpha^0 - T \frac{\partial \mu_\alpha^0}{\partial T} |_{T, x_{\delta \neq \alpha, \beta}} = H_\alpha^0 = C_p \alpha T).$$

$$\vec{J}_\alpha = - \sum_{\beta=1}^N \frac{\mathcal{R} T a_{\alpha\beta}}{W_\beta} \left(\frac{\nabla x_\beta}{x_\beta} + \frac{\nabla p^0}{p^0} + \frac{1}{\mathcal{R} T} \nabla (\mu_\beta^I - \nabla \cdot (\rho \lambda_\beta \nabla n_\beta)) \right)$$

$$\vec{q} = -k \nabla T + \sum_{\alpha=1}^N H_\alpha \frac{\vec{J}_\alpha}{W_\alpha} + \sum_{\alpha=1}^N \mu_\alpha^I \frac{\vec{J}_\alpha}{W_\alpha} - \sum_{\alpha=1}^N \nabla \cdot (\rho \lambda_\alpha \nabla n_\alpha) \frac{\vec{J}_\alpha}{W_\alpha}$$

$$+ \sum_{\alpha=1}^N \rho \lambda_\alpha \nabla n_\alpha \left(n_\alpha \nabla \cdot \vec{U} + \nabla \cdot \frac{\vec{J}_\alpha}{W_\alpha} - \dot{\omega}_\alpha \right)$$

$$\underline{\underline{\sigma^r}} = - \left(p^0 + p^I + \frac{\rho^2}{2} \sum_{\alpha=1}^N \frac{\partial \lambda_\alpha}{\partial \rho} |\nabla n_\alpha|^2 - \sum_{\alpha=1}^N n_\alpha \nabla \cdot (\rho \lambda_\alpha \nabla n_\alpha) \right) \underline{\underline{I}} - \sum_{\alpha=1}^N \rho \lambda_\alpha (\nabla n_\alpha \otimes \nabla n_\alpha)$$

Further simplification of the general transport terms

❖ Assume:

- 1) Binary flow ($N=2$)
- 2) Assume a double-well potential; take the order parameter $\epsilon = 2Y_1 - 1$; $\hat{F}^I = A^I \frac{\sigma}{4} (\epsilon^2 - 1)^2$; Take $\sigma = \frac{\mathcal{R}T}{W'}$, where $W' = \frac{2 W_1 W_2}{W_1 + W_2}$ is the mixture molar mass at $Y_1 = 0.5$.
- 3) Assume $\lambda_\alpha = A_\alpha \mathcal{R} T W_\alpha^2 / (W' \rho^2)$.
- 4) Mobility is given by $a_{11} = \frac{\rho}{\mathcal{R}T} \frac{Y_1 Y_2 W_1 W_2}{W} \mathcal{D}_{12}$ so that Fick's law can be formally satisfied.
- 5) Take $A_1 = A_2$; $L^2 = 2A_1/A_I$ (dimension $[\text{m}^2]$);

Final expressions for the compressible transport terms

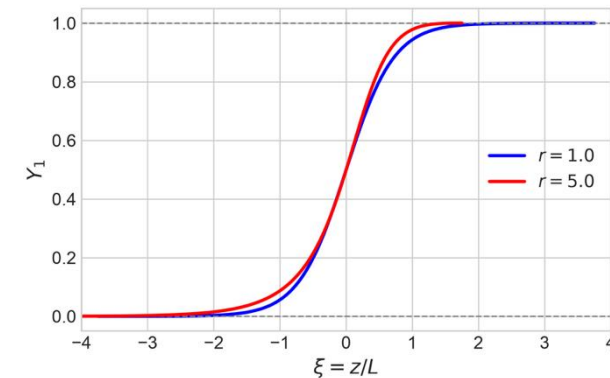
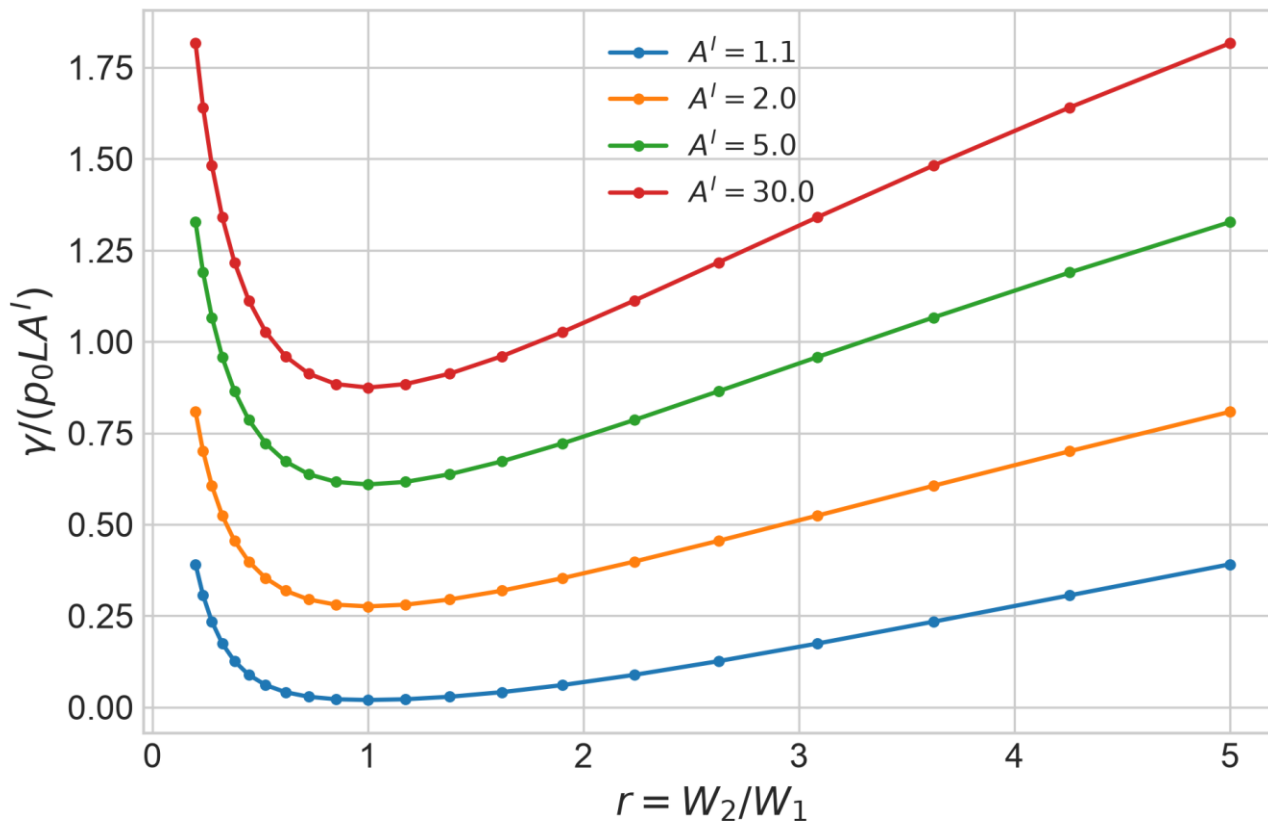
$$\begin{aligned} \vec{J}_1 &= -\rho \mathcal{D}_{12} \left(\nabla Y_1 + Y_1 Y_2 \frac{W_2 - W_1}{W} \frac{\nabla p^0}{p^0} \right. \\ &\quad \left. + \frac{4 A^I W_2 + W_1}{\mathcal{R}T} Y_1 Y_2 \nabla \left[\mathcal{R}T Y_1 Y_2 (Y_2 - Y_1) - \frac{L^2}{8} \nabla \cdot \left[\mathcal{R}T \left(\frac{\nabla(\rho Y_1)}{\rho} - \frac{1}{2} \frac{\nabla(\rho)}{\rho} \right) \right] \right] \right) \end{aligned}$$

$$\begin{aligned} \vec{q} &= -k \nabla T + \sum_{\alpha=1}^2 c_{p\alpha} T \vec{J}_\alpha + 4 A^I \frac{W_2 + W_1}{W_1 W_2} \left[\mathcal{R}T Y_1 Y_2 (Y_2 - Y_1) - \frac{L^2}{8} \nabla \cdot \left[\mathcal{R}T \left(\frac{\nabla(\rho Y_1)}{\rho} - \frac{1}{2} \frac{\nabla(\rho)}{\rho} \right) \right] \right] \vec{J}_1 \\ &\quad + A^I \frac{L^2}{4} \mathcal{R}T \frac{W_2 + W_1}{W_1 W_2} \sum_{\alpha=1}^2 \frac{\nabla(\rho Y_\alpha)}{\rho} (\rho Y_\alpha \nabla \cdot \vec{U} + \nabla \cdot \vec{J}_\alpha - \dot{\omega}_\alpha W_\alpha) \end{aligned}$$

$$\begin{aligned} \underline{\underline{\sigma^r}} &= - \left(p^0 - A^I \frac{L^2}{4} \rho \mathcal{R}T \frac{W_2 + W_1}{W_1 W_2} \left[\sum_{\alpha=1}^2 \left| \frac{\nabla(\rho Y_\alpha)}{\rho} \right|^2 + Y_\alpha \nabla \cdot \left(\frac{\nabla(\rho Y_\alpha)}{\rho} \right) + \frac{1}{\mathcal{R}T} \nabla(\mathcal{R}T) \cdot \frac{\nabla(\rho Y_\alpha)}{\rho} \right] \right) \underline{\underline{I}} \\ &\quad - A^I \frac{L^2}{4} \rho \mathcal{R}T \frac{W_2 + W_1}{W_1 W_2} \sum_{\alpha=1}^2 \frac{\nabla(\rho Y_\alpha)}{\rho} \otimes \frac{\nabla(\rho Y_\alpha)}{\rho} \end{aligned}$$

Equilibrium solutions for the compressible flux

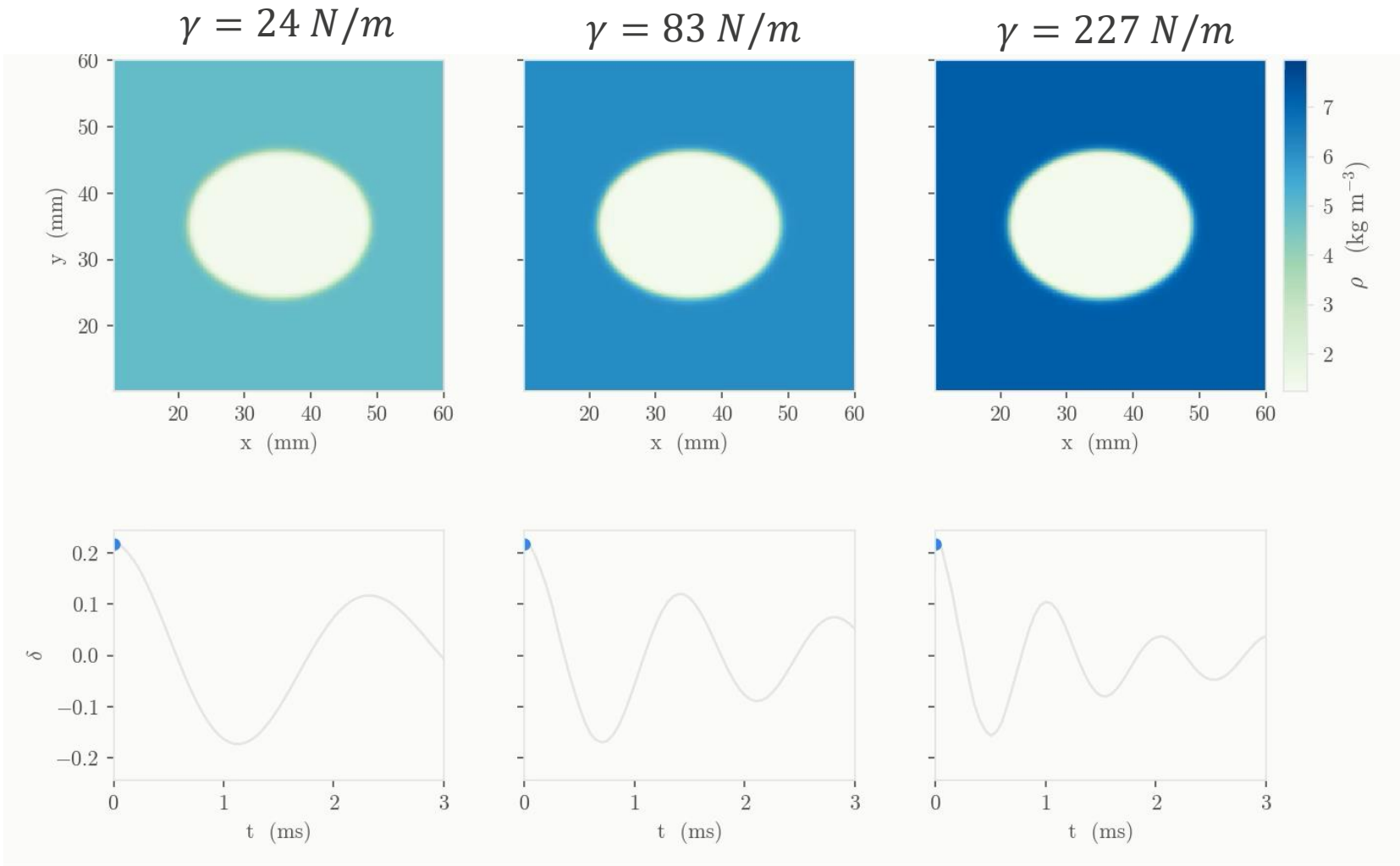
Take $\vec{J}_1 = 0$ to obtain the mass fraction solution and integrate the free energy over the interface to obtain the surface tension.



- Slightly smoother profiles and larger surface tension at $r \neq 1$.
- $r = 1$ and $A^I \rightarrow \infty$ recovers the incompressible uniform density limit of $2\sqrt{2}/3$.

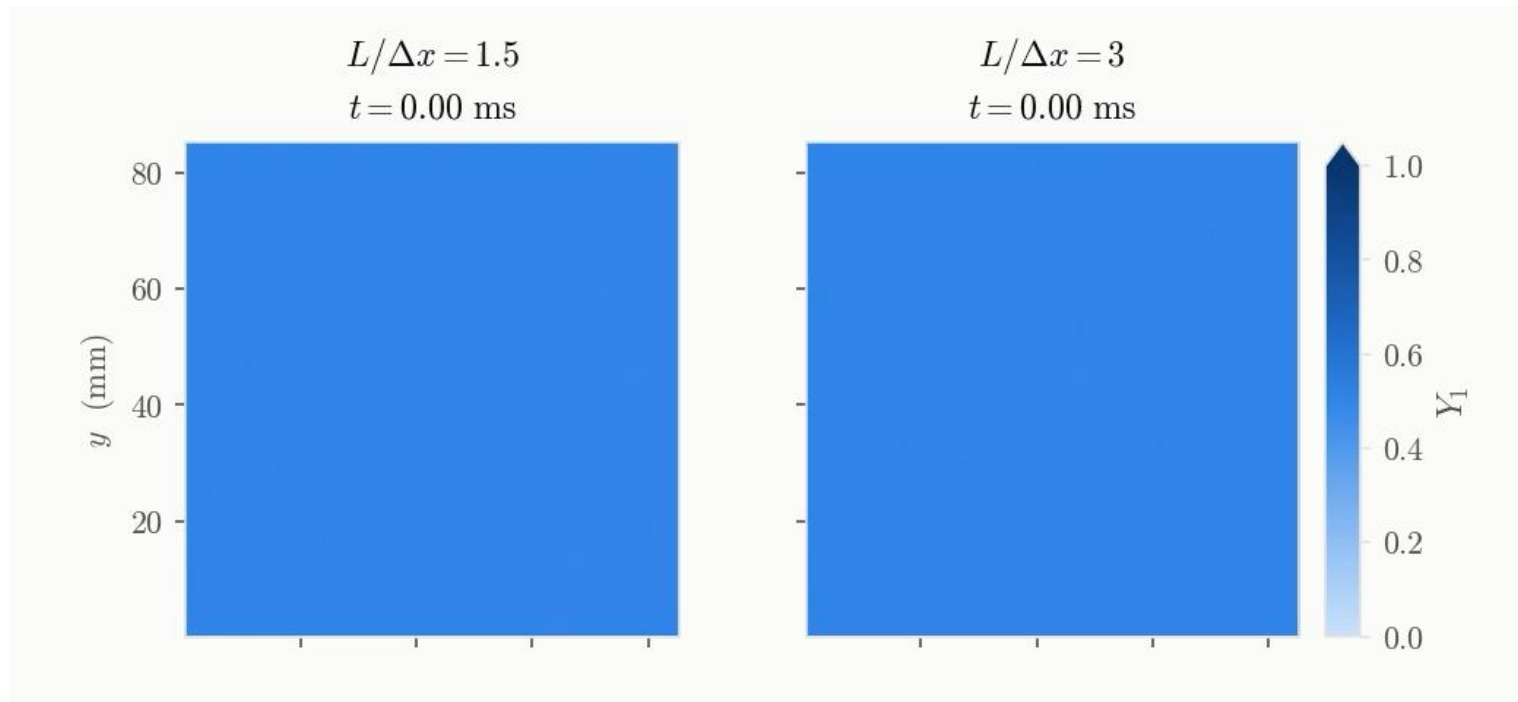
Bubble oscillations at $r = 10$ with different surface tension values

Oscillations are consistent with Fyfe et al, JCP 1987, Garrick et al, JCP 2017.



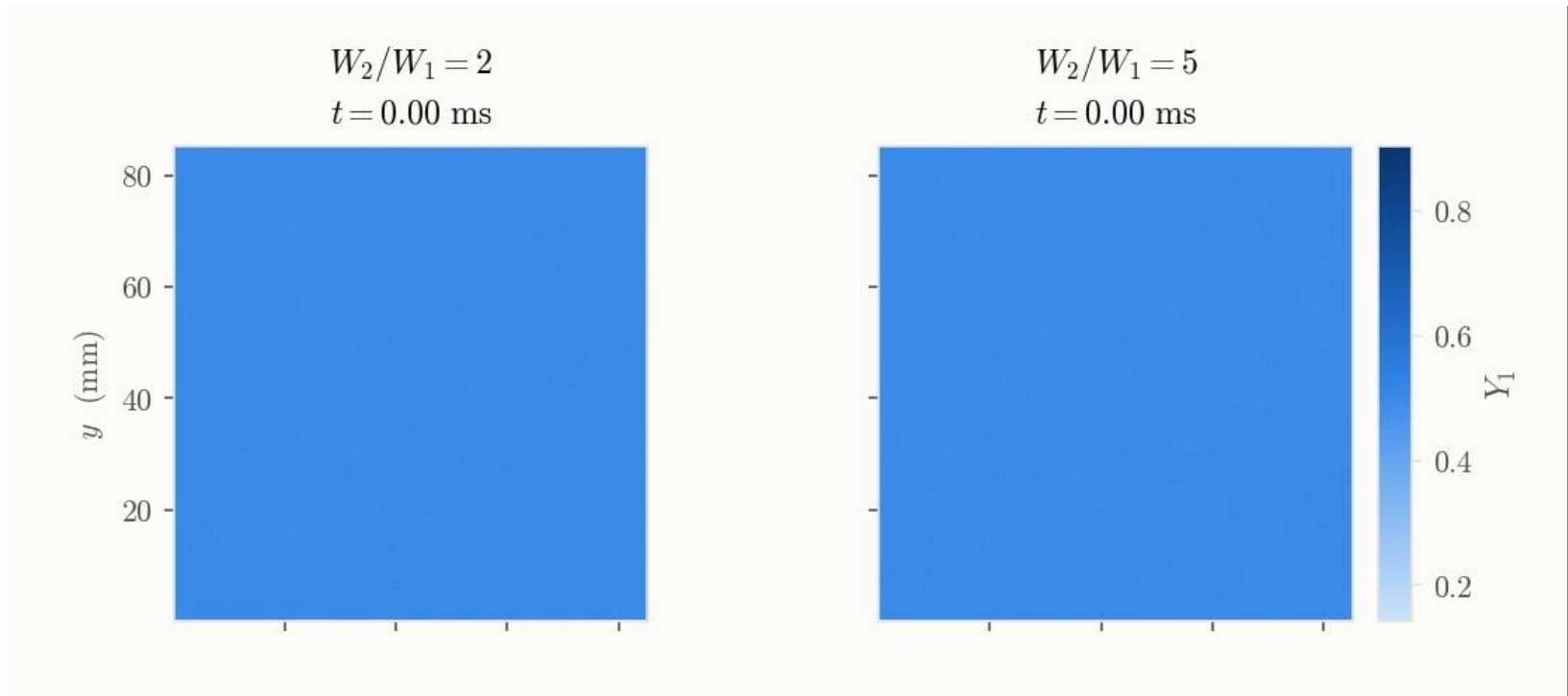
Separation from a mixture with small perturbations (spinodal decomposition)

- Simulations recover the dispersion relation for a small amplitude growth
- Effect of interface width:



Separation from a mixture with small perturbations (spinodal decomposition)

- Effect of density ratio:



Incompressible Generalized Cahn-Hilliard Navier-Stokes Equations (IGCHNS)

- The incompressible limit is obtained as the infinite speed of sound limit. Here, we choose $T \rightarrow \infty, \frac{\partial p_0}{\partial T} \rightarrow \text{const}$ (Livescu, Phil. Trans. R. Soc. A, 2013).
- The interfacial Helmholtz free energy, chemical potential and all coefficients lose their temperature dependence.
- The Continuity and Momentum equations maintain the same formulations as for compressible flows.
- The Internal energy equation and Species mass fraction equations are reduced to an identical expression for the velocity divergence:

$$\nabla \cdot \vec{u} = -\frac{1}{\partial p / \partial \rho|_{T, Y_\alpha}} \sum_{\alpha=1}^N \frac{\partial p}{\partial Y_\alpha} |_{\rho, T} \nabla \cdot \vec{J}_\alpha$$

- Extensive set of tests, including drop deformation in shear flow, head-on collision, immiscible Rayleigh-Taylor instability.

Late time multi-mode incompressible Rayleigh-Taylor instability with 3:1 density ratio

Grid-Convergence check

Scale separation is defined as $SS = \frac{\eta_K}{L}$

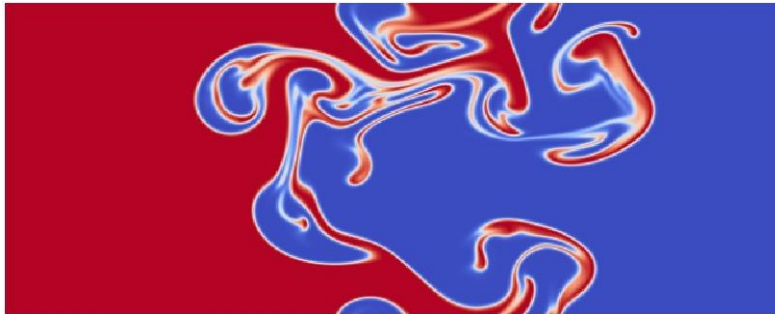
η_K : Kolmogorov scale
1:1

R0: 640 × 256, R1: 1280 × 512

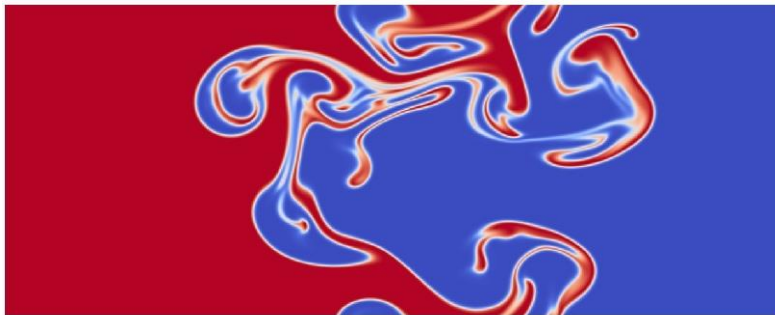
R2: 2560 × 1024, R3: 5120 × 2048

SS=0.5

R0

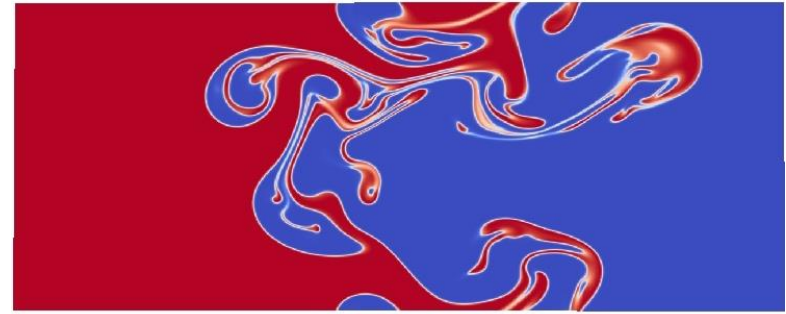


R1

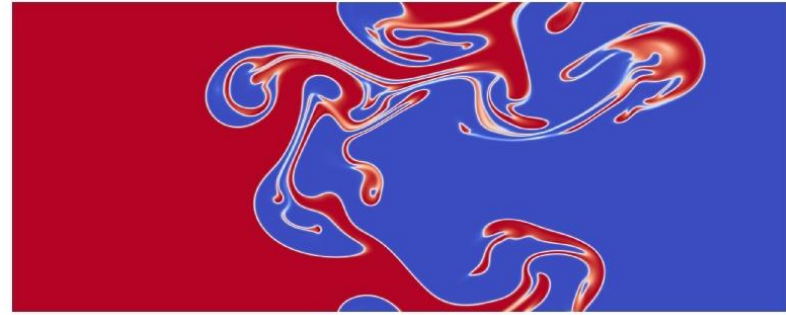


SS=1.0

R2



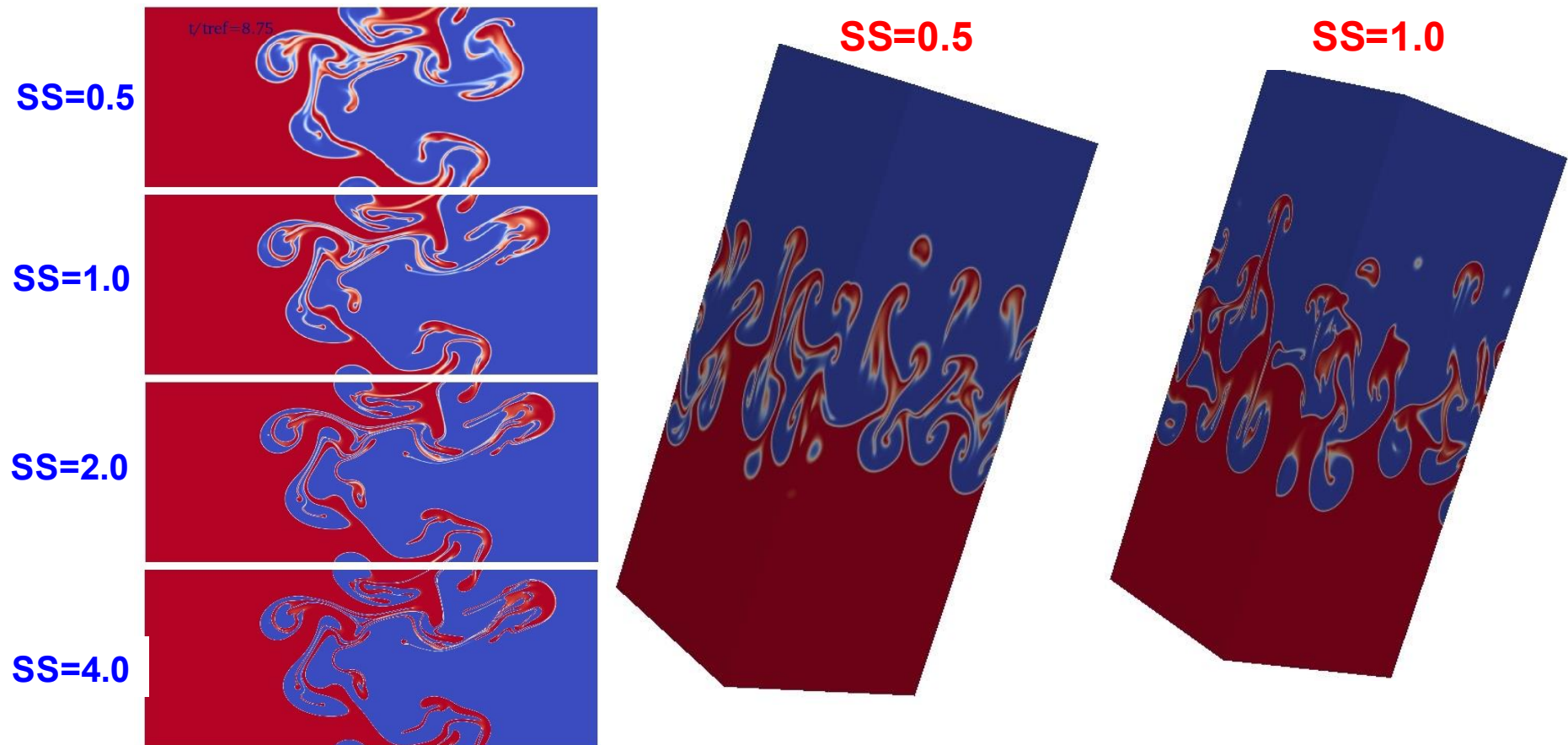
R3



Late time multi-mode incompressible Rayleigh-Taylor instability with 3:1 density ratio

Results Convergence Fast with Scale-Separation, $SS = \frac{\eta_K}{L}$

(all results presented are also grid converged)



Summary

- The compressible generalized Cahn-Hilliard Navier-Stokes (CGCHNS) equations were derived from basic thermodynamics for multi-component ($N \geq 2$) flows.
- The incompressible generalized Cahn-Hilliard Navier-Stokes (IGCHNS) equations were rigorously derived as the incompressible limit of CGCHNS. The equations can address fluids with arbitrary density ratios.
- The Cahn-Hilliard Navier-Stokes equations can naturally handle complex interface deformation, including merging and breaking.
- Extensive tests have been performed for both the fully compressible case and variable density incompressible limit.
- Late-time simulations of the incompressible variable density Rayleigh-Taylor instability show that the results converge with both grid resolution and scale separation, with a ratio of $\frac{\eta_K}{L} \approx 2 \div 4$ being enough in most cases for convergence.